Crystal Chiu

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Education

B.S. in Computer Science March 2025
University of California, Irvine

Experience

Software Engineer — (AWS, Docker, Terraform, Laravel, Vue.js, React.js, Next.js, MySQL, Redis) Jul. 2025 - Present
The Alliance Group Enterprise *Santa Ana, CA*

- Re-architected legacy infrastructure to AWS using Terraform to managing cloud resources and Docker to to containerize services, migrating application data and assets without loss or disruption, and improving deployment flexibility to support long-term scalability
- Implemented an image-based CI/CD pipeline using GitHub Actions with separate workflows for testing, image builds, and environment-specific deployments, replacing manual processes, improving reliability, and increasing production uptime from 90% to 99.9%
- Designed and built an internal timesheet system optimized for civil engineers' workflows, reducing admin overhead for payroll and prime contractor invoicing, replacing a third-party vendor (>$100/month savings), and unlocking live labor cost metrics and KPIs
- Led a progressive migration from a legacy server-rendered Blade application to a Vue.js single-page architecture, redesigning core workflows and user interfaces to improve usability, streamline interaction patterns, and support a more intuitive end-user experience, while transitioning backend services into RESTful APIs
- Improved the performance and scalability of backend APIs by addressing inefficient data access patterns and introducing caching, reducing response times on high load endpoints from 5-7s to 400-600ms and easing database load
- Established strict and structured AI development workflows by creating domain-specific context files to guide AI-assisted development, improving output consistency and reducing redundant prompting and rework, resulting in an overall 25-35% reduction in token usage

Student Software Developer — (Laravel, Vue.js, Cypress, MySQL) Mar. 2023 - Dec. 2024
UCI Student Affairs IT *Irvine, CA*

- Worked on full-stack applications serving 10k+ students and staff, handling feature development, bug fixes, refactoring, and iterative improvements using Agile and TDD practices under quarterly release timelines
- Mentored 3 new developers in a large, long-running UCI Commencement application (10+ years in production), helping them familiarize with team conventions, onboard onto unfamiliar frameworks, and debug complex issues
- Assisted in building a custom, dynamic seating chart tool for commencement organizers, enabling automated seat assignments and speeding up a previously tedious planning workflow by weeks
- Collaborated with supervisor in summer planning sessions for projects by collaborating with stakeholders, writing design docs, designing and presenting feature mockups, and translating high-level requirements into tickets for the student team

Projects

Popmart Restock Monitor | *Javascript (Node.js), MongoDB, Discord API*

- Developed and deployed an API proxy scraper on an AWS EC2 instance and a Raspberry Pi to track Popmart's "Labubu" product restocks and product changes (price, availability, etc.), sending real-time Discord alerts to assist users in purchasing high-demand items with unpredictable availability
- Created multiple modes (Regular, Snooze, Throttle) for polite scraping to reduce unnecessary network traffic based on likelihood of restock time and priority items in the product line
- Optimized Puppeteer scraping of JavaScript-heavy content, reducing page load and scrape time from 20s to 5s avg. per page
- Implemented robust error handling to manage network failures, page errors, etc. while monitoring program health with PM2

Movie Catalogue Website & Android App | *Java Servlets, Maven, Apache Tomcat, MySQL*

- Co-developed and deployed a web and Android app managing several GB of movie data, enabling users to upload, search (by genre/title), view, and simulate purchases
- Wrote an efficient data import script using SAX parsing to read large XML files and upload clean data to the production MySQL database in under 10 minutes
- Implemented CAPTCHA, secure authentication, session tracking, connection pooling, and load balancing across 2 Micro EC2 instances for seamless traffic flow and improved personalization

Technical Skills

Languages: Java, Python, JavaScript, C++, PHP, SQL
Frameworks: Vue.js, React.js, Node.js, Flask, Laravel
Databases: MySQL, PostgreSQL, SQL++, Redis
AI & Applied Machine Learning: Prompt Design, AI Usage Governance, and ML Fundamentals (Classification, Regression, Reinforcement Learning Concepts)
Tooling & Infrastructure: Git/GitHub, Docker, AWS, CI/CD (GitHub Actions), Terraform, Jira, Figma